

Robotics and Communications Systems Technology

RCET 2251: Systems Analog & Digital Theory

Day One Test Prep

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1. State Ohm's Law.

Ohm's Law states that voltage is equal to current multiplied by resistance ($V = IR$).

2. State the electrical power equation.

Electrical power is equal to voltage multiplied by current ($P = VI$).

3. State Kirchhoff's Voltage Law (KVL).

The algebraic sum of all voltages around a closed loop is zero.

4. State Kirchhoff's Current Law (KCL).

The algebraic sum of all currents entering and leaving a node is zero.

5. State Thevenin's Theorem.

Any linear two-terminal circuit can be replaced by an equivalent circuit consisting of a single voltage source in series with an equivalent resistance.

6. State Norton's Theorem.

Any linear two-terminal circuit can be replaced by an equivalent circuit consisting of a single current source in parallel with an equivalent resistance.

7. See Figure 1. Resistor Circuit, If $V_{DC} = 50V$, $R1 = 1K\Omega$, $R2 = 2K\Omega$, $R3 = 3K\Omega$, and the current through the load, $I_{RL} = 20mA$. Find the total circuit current I_{Total} .

$I_{Total} = 25mA$

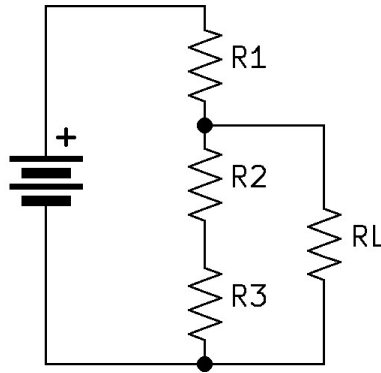


Figure 1: Resistor Circuit

8. See Figure 2. Resistor Circuit Analysis, If $V_{DC} = 6V$, $R1 = 1K\Omega$, $R2 = 4K\Omega$, and $R3 = 4K\Omega$, what is V_{R3}

$V_{R3} = 4V$

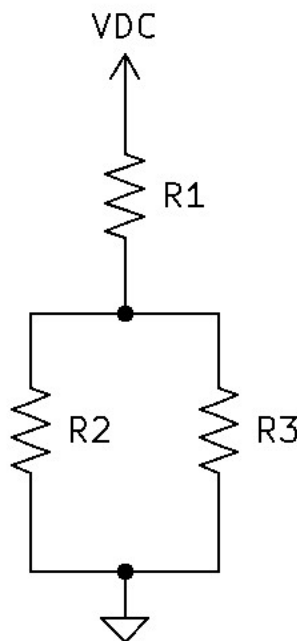


Figure 2: Resistor Circuit Analysis

9. See Figure 3. Resistor Circuit Analysis, If $V_{DC} = 12V$, $R_1 = 1.2K\Omega$, $R_2 = 4.7K\Omega$, $R_3 = 2K\Omega$, and $R_4 = 2.7K\Omega$ Find: R_{Thev} and V_{Thev} .

$$R_{Thev} = 2.46K\Omega \text{ and } V_{Thev} = 3.1V$$

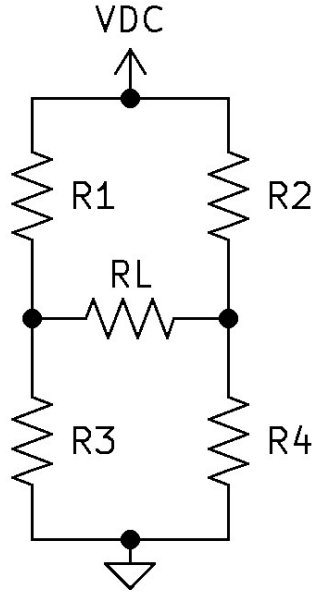


Figure 3: Resistor Circuit Analysis

10. See Figure 4. ABCD Circuit 1, If $V_{DC1} = 12V$, $V_{DC2} = 14V$, $R_1 = 20\Omega$, $R_2 = 40\Omega$, and $R_3 = 60\Omega$

Find: V_{AB} , V_{BC} , V_{CD} , V_{AD}

Find: V_{BA} , V_{CB} , V_{DC} , V_{DA}

Find: V_A , V_B , V_C , V_D

$$V_{AB} = 4.333V, V_{BC} = 8.666V, V_{CD} = 13V, V_{AD} = 26V$$

$$V_{BA} = -4.333V, V_{CB} = -8.666V, V_{DC} = -13V, V_{DA} = -26V$$

$$V_A = 12V, V_B = 7.667V, V_C = -1V, V_D = -14V$$

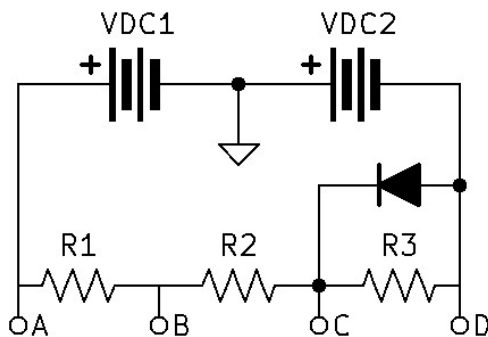


Figure 4: ABCD Circuit 1

11. See Figure 5. ABCD Circuit 2, If $V_{DC1} = 12V$, $V_{DC2} = 14V$, $R1 = 20\Omega$, $R2 = 40\Omega$, and $R3 = 60\Omega$

Find: V_{AB} , V_{BC} , V_{CD} , V_{AD}

Find: V_{BA} , V_{CB} , V_{DC} , V_{DA}

Find: V_A , V_B , V_C , V_D

$$V_{AB} = 8.433V, V_{BC} = 16.867V, V_{CD} = 0.7V, V_{AD} = 26V$$

$$V_{BA} = -8.433V, V_{CB} = -16.867V, V_{DC} = -0.7V, V_{DA} = -26V$$

$$V_A = 12V, V_B = 3.567V, V_C = -13.3V, V_D = -14V$$

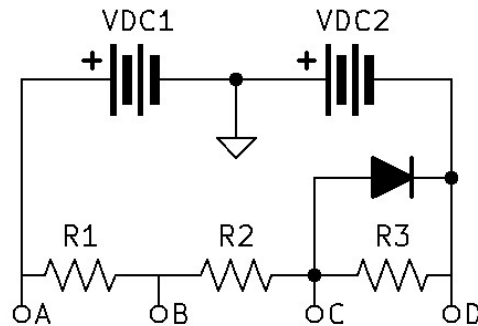


Figure 5: ABCD Circuit 2

12. Write the formula used to calculate capacitive reactance.

$$X_C = \frac{1}{2\pi FC}$$

13. Write the formula used to calculate inductive reactance.

$$X_L = 2\pi FL$$

14. Write the formula used to calculate frequency resonance in an RCL circuit.

$$F_R = \frac{1}{2\pi\sqrt{LC}}$$

15. State the definition of impedance.

Impedance is the total opposition to alternating current in a circuit and consists of both resistance and reactance, measured in ohms (Ω).

16. See Figure 6. Filter Identification 1, Identify the following frequency response (High-Pass or Low-Pass) for each circuit.

- A. = High-Pass
 B. = Low-Pass
 C. = Low-Pass
 D. = High-Pass

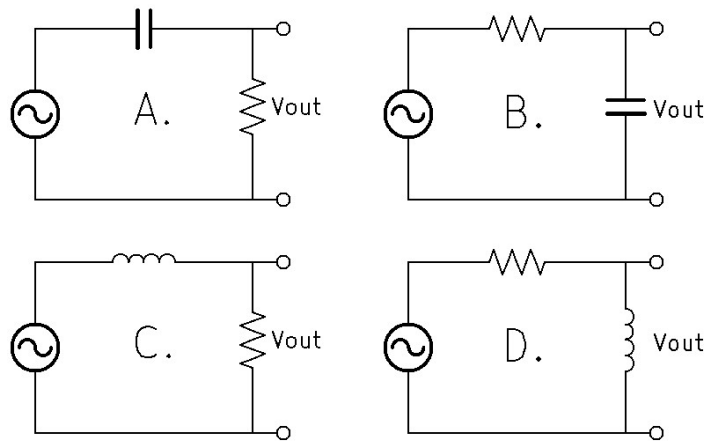


Figure 6: Filter Identification 1

17. See Figure 7. Filter Identification 2, Identify the following frequency response (Band-Pass or Band-Stop) for each circuit.

- A. = Band-Pass
 B. = Band-Stop
 C. = Band-Stop
 D. = Band-Pass

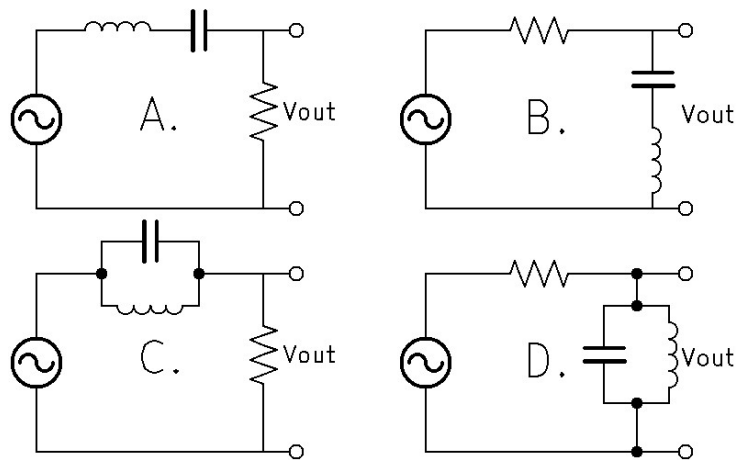


Figure 7: Filter Identification 2

18. See Figure 8. RC Circuit, If $V_{IN} = 5vp$, $C = 0.047\mu F$, and $R = 4.7K\Omega$, Find the critical frequency, the resistor, and the capacitor voltages at the critical frequency.

$$F_C = 720.5hz$$

$$V_R = 3.535vp, \angle 45^\circ$$

$$V_C = 3.535vp, \angle -45^\circ$$

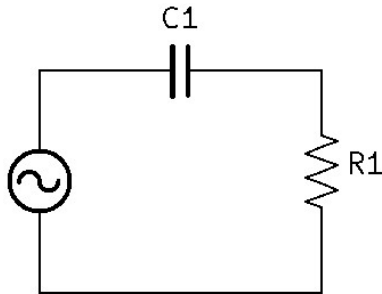


Figure 8: RC Circuit

19. See Figure 9. RCL Circuit, If $R1 = 1K\Omega$, $XC1 = 3.5K\Omega$, $R2 = 10K\Omega$, $XL1 = 4K\Omega$, $R3 = 20K\Omega$, and $XC2 = 10K\Omega$, find the equivalent series circuit resistance in Polar and Rectangular form.

$$Z_{total}(Polar) = 9.196K\Omega \angle -16.437^\circ$$

$$Z_{total}(Rec) = 8.82K\Omega, -j2.603K\Omega$$

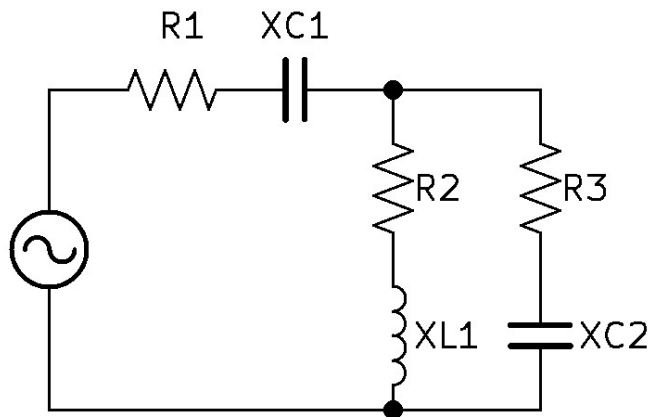


Figure 9: RCL Circuit

20. See Figure 10. RCL Circuit, If $V_{IN} = 10vp$, $Freq_{Gen} = 1Khz$, $R1 = 4.3K\Omega$, $R2 = 10K\Omega$, $R3 = 4.7K\Omega$, $C1 = 20nF$, $C2 = 8nF$, and $L = 3.5H$, find V_{R1} , V_{R2} , and V_{R3} , include the phase angle in reference to V_{Gen} .

$$V_{R1} = 1.064vp \angle 27.949^\circ$$

$$V_{R2} = 3.407vp \angle -56.876^\circ$$

$$V_{R3} = 1.892vp \angle 85.378^\circ$$

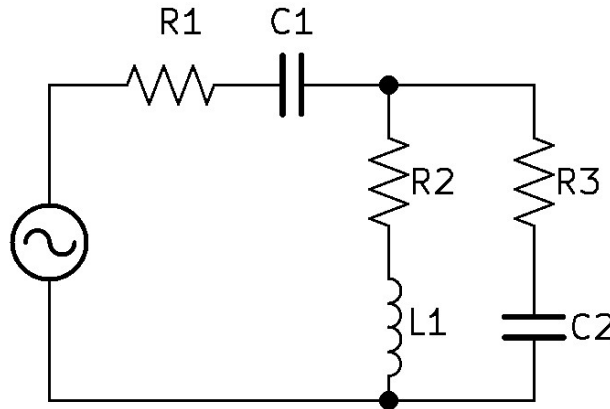


Figure 10: RCL Circuit

21. Write the equation for the capacitor charge formula.

$$V_C = V_{fin} - (V_{fin} - V_{int})e^{-\frac{t}{\tau}}$$

22. See Figure 11. Capacitor Charge Circuit, If $V_{DC} = 10V$, $R1 = 1K\Omega$, and $C1 = 1\mu F$, Find the capacitor and resistor voltages at the following times: $500\mu Sec$, $1mSec$, $1.5mSec$, $2mSec$, $2.5mSec$, and $3mSec$.

$$\text{At } 500\mu Sec, V_C = 3.935V, V_R = 6.065V$$

$$\text{At } 1.0mSec, V_C = 6.321V, V_R = 3.679V$$

$$\text{At } 1.5mSec, V_C = 7.769V, V_R = 2.231V$$

$$\text{At } 2.0mSec, V_C = 8.647V, V_R = 1.353V$$

$$\text{At } 2.5mSec, V_C = 9.179V, V_R = 0.821V$$

$$\text{At } 3.0mSec, V_C = 9.502V, V_R = 0.498V$$

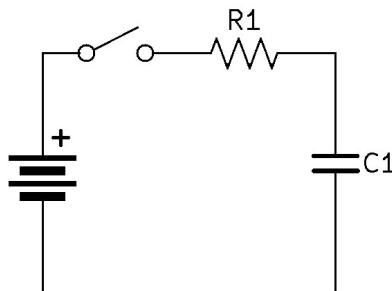


Figure 11: Capacitor Charge Circuit

23. See Figure 12. Pi-Filtered Rectifier, If $AC = 120V_{RMS}$, the turns ratio of the transformer is 4:1, $C1 = 22\mu F$, $R1 = 560\Omega$, $C2 = 47\mu F$, and $RL = 2.7K\Omega$, Find: V_{max_A} , V_{min_A} , V_{DC_A} , V_{rpp_A} , V_{DC_B} , V_{rpp_B} , and the output ripple factor ($r\%$).

$$\begin{aligned} V_{max_A} &= 41.03vp \\ V_{min_A} &= 36.53vp \\ V_{DC} &= 38.78V \\ V_{rpp} &= 4.5V_{pp} \\ V_{DC_B} &= 32.12V \\ V_{rpp_B} &= 225mV_{pp} \\ r\% &= 0.202\% \end{aligned}$$

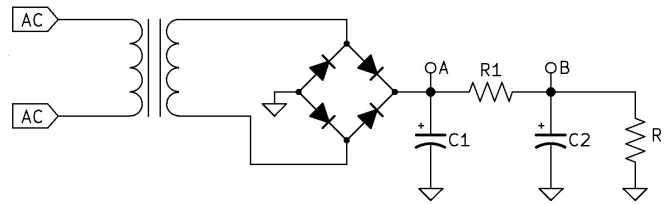


Figure 12: Pi-Filtered Rectifier

24. A full-wave rectifier has a $200\mu f$ capacitor and produces a ripple voltage that varies from 18V to 16.5V. If a 120VAC 60Hz source supplies the input, what is the value of the load resistor (RL)?

$$RL = 478.86\Omega$$

25. See Figure 13. Bipolar Transistor Biasing Configurations, Identify the following Bipolar Transistor Biasing Configurations.

- Base Bias
- Universal Bias Or Voltage Divider Bias
- Collector Feedback Bias
- Emmitter Bias Or Two Power Supply Bias
- Emmitter Bias Or Two Power Supply Bias

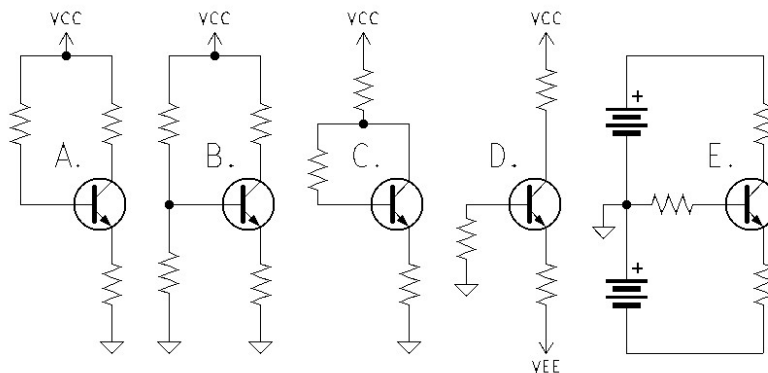


Figure 13: Bipolar Transistor Biasing Configurations

26. List Three Bipolar Junction Transistor Amplifier Configurations and their Characteristics.

Common Emitter: Input on the Base, Output on the Collector, Good Voltage Gain (ΔV), Good Current Gain (ΔI), Great Power Gain (ΔP), High input impedance (Z_{IN}), and High output impedance (Z_{OUT}), the output waveform is 180° out of phase with the input.

Common Collector: Input on the Base, Output on the Emitter, less than unity Voltage Gain ($\Delta V \approx 1$), Good Current Gain (ΔI), Good Power Gain (ΔP), High input impedance (Z_{IN}), and Low output impedance (Z_{OUT}), the output waveform is in phase with the input.

Common Base: Input on the Emitter, Output on the Collector, Good Voltage Gain (ΔV), less than unity Current Gain ($\Delta I \approx 1$), Good Power Gain (ΔP), Low input impedance (Z_{IN}), and High output impedance (Z_{OUT}), the output waveform is in phase with the input.

27. See Figure 14. BJT Amplifier, If $V_{CC} = 18V$, $R1 = 910K\Omega$, $R2 = 3K\Omega$, $R3 = 200\Omega$, $R4 = 800\Omega$, $R_L = 3K\Omega$, and $\beta = 125$. At mid-band find: V_B , V_C , V_E , Z_{IN} , Z_{OUT} , ΔV , ΔI , ΔP ,

$$V_B = 2.804V, V_C = 11.739V, V_E = 2.104V$$

$$Z_{iN} = 25.993K\Omega, Z_{OUT} = 3K\Omega$$

$$\Delta V = -7.005, \Delta I = -121.387, \Delta P = 850.316$$

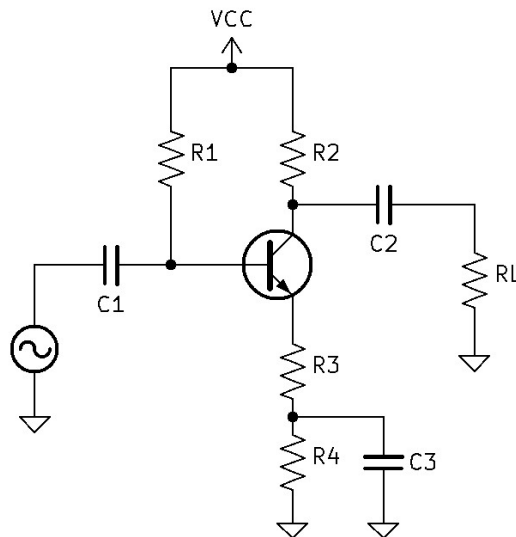


Figure 14: BJT Amplifier

28. See Figure 15. BJT Amplifier, If $V_{CC} = 15V$, $R_{gen} = 50\Omega$, $R_1 = 90K\Omega$, $R_2 = 330\Omega$, $R_3 = 120\Omega$, $R_L = 120\Omega$, and $\beta = 100$. At mid-band find: Z_{IN} , Z_{OUT} , and gains,

$$Z_{iN} = 5.84K\Omega, Z_{OUT} = 2.29\Omega$$

$$\Delta V = 0.970, \Delta I = 47.207, \Delta P = 45.79$$

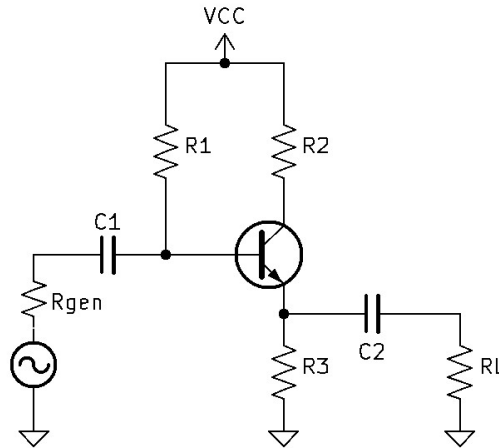


Figure 15: BJT Amplifier

29. See Figure 16. BJT Amplifier, If $V_{CC} = 18V$, $R_{gen} = 75\Omega$, $R_1 = 33K\Omega$, $R_2 = 5.6K\Omega$, $R_3 = 1.5K\Omega$, $R_4 = 100\Omega$, $R_5 = 290\Omega$, $R_L = 1.5K\Omega$, and $\beta = 200$. At mid-band find: Z_{IN} , Z_{OUT} , and gains,

$$Z_{iN} = 77.427\Omega, Z_{OUT} = 1.5K\Omega$$

$$\Delta V = 7.065, \Delta I = 0.365, \Delta P = 2.555$$

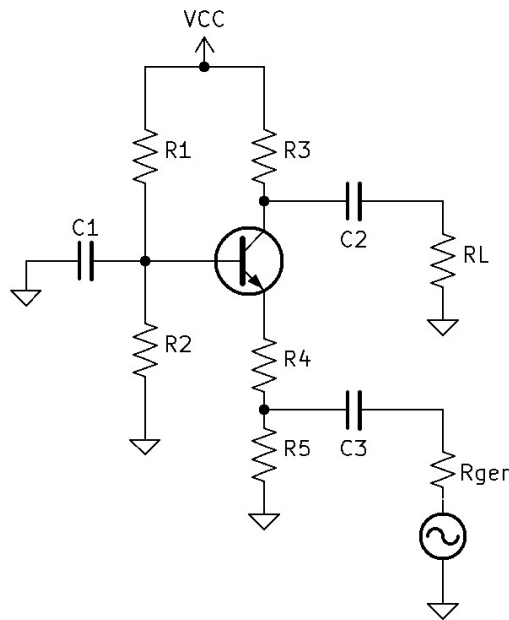


Figure 16: BJT Amplifier

30. List the four Characteristics of an Ideal Operational Amplifier.

- ∞ Gain
- ∞ input impedance Z_{IN}
- ∞ Frequency Response – Bandwidth
- 0 output impedance Z_{OUT}

31. List the two rules for calculating Operational Amplifiers.

- No Signal Current into or out of the inputs.
- The output will do whatever it can to make the two input voltages the same.

32. List the Six Steps for calculating Operational Amplifiers.

1. Find V_+
2. w/feedback, $V_- = V_+$
3. Find V_{RI}
4. Find I_{RI} , $I_{RF} = I_{RI}$
5. Find V_{RF}
6. Kirchhoff to find V_{OUT}

33. See Figure 17. Operational Amplifier, If $V_{IN} = 1vp$, $RI = 1K\Omega$, and $RF = 10K\Omega$, find: ΔV and R_B .

- $\Delta V = -10$
- $R_B = 909\Omega$

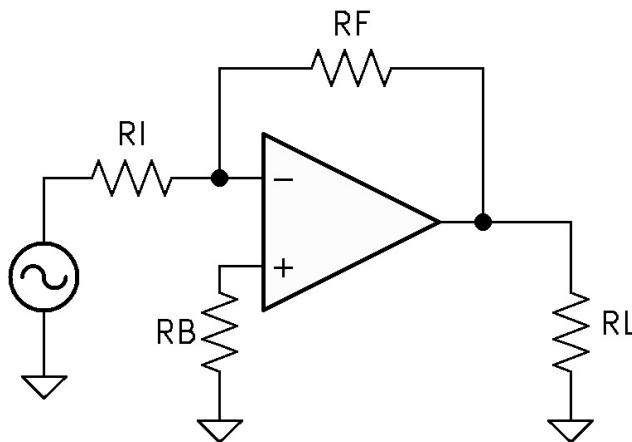


Figure 17: Operational Amplifier

34. See Figure 18. Operational Amplifier, If $V_{IN} = 1vp$, $R_I = 1K\Omega$, and $R_F = 10K\Omega$, find: ΔV and R_B .

$$\Delta V = 11$$

$$R_B = 909\Omega$$

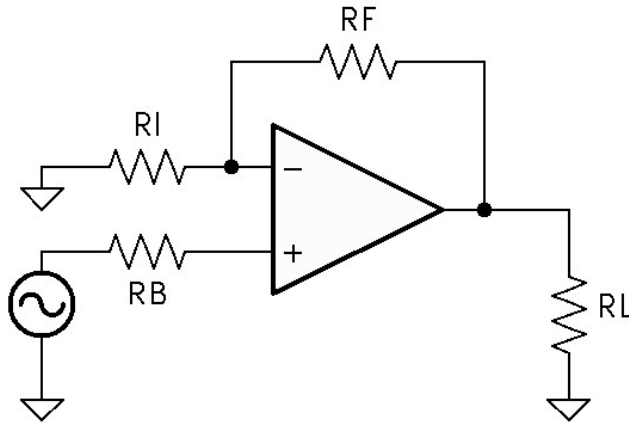


Figure 18: Operational Amplifier

35. See Figure 19. Operational Amplifier, If $V_1 = 900mV$, $V_2 = 600mV$, $V_3 = 300mV$, $R_1 = 2K\Omega$, $R_2 = 1.5K\Omega$, $R_3 = 1K\Omega$, and $R_F = 6K\Omega$, find: V_{OUT} and R_B .

$$V_{OUT} = -6.9V$$

$$R_B = 429\Omega$$

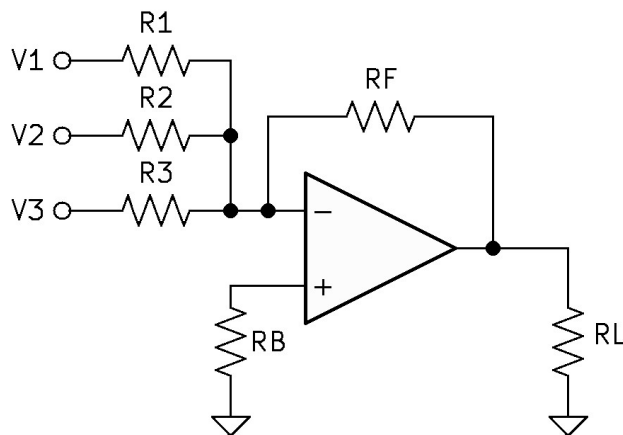


Figure 19: Operational Amplifier

36. See Figure 20. Operational Amplifier, If $V_1 = 4.5V$, $V_2 = 2V$, R_1 - R_4 (all resistors) = $1.5K\Omega$, find: V_{OUT} .

$$V_{OUT} = -2.5V$$

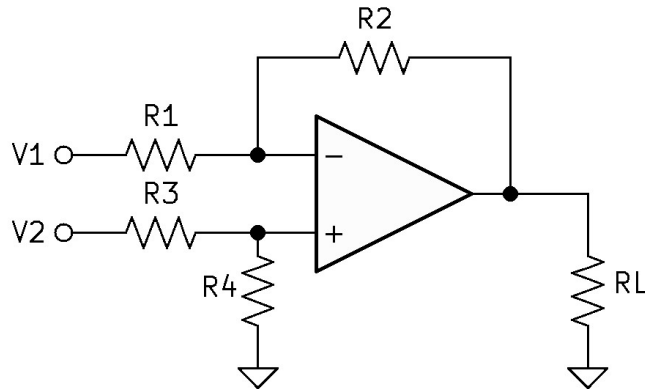


Figure 20: Operational Amplifier

37. See Figure 20. Operational Amplifier. See the previous question, find V_{OUT} , when the values of R_2 and R_4 are changed to $6K\Omega$. find: V_{OUT} .

$$V_{OUT} = -10V$$

38. See Figure 21. Operational Amplifier, If $V_1 = 2V$, find: V_{OUT} .

$$V_{OUT} = 2V$$

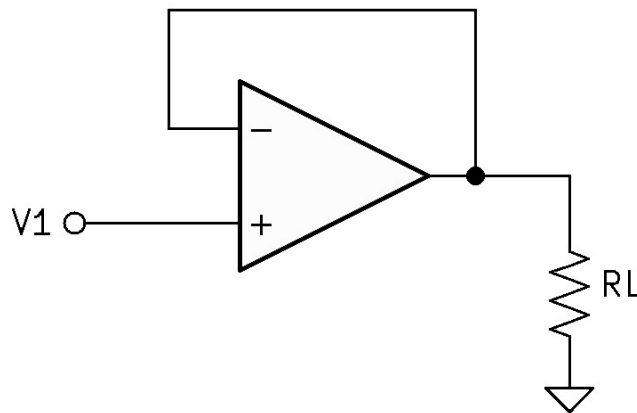


Figure 21: Operational Amplifier

39. Convert 37_{10} and 221_{10} to binary.

$$37_{10} = 100101_2$$

$$221_{10} = 11011101_2$$

40. Convert 63_8 , $D9_{16}$, and 10011001_2 to decimal.

$$63_8 = 51_{10}$$

$$D9_{16} = 217_{10}$$

$$10011001_2 = 153_{10}$$

41. List the following TTL values: $I_{OutHMax}$, $I_{OutLMax}$, $V_{OutHTyp.}$, and $V_{OutLTyp.}$

$$I_{out_HIGH}Max = 400\mu A$$

$$I_{out_LOW}Max = 16mA$$

$$V_{out_HIGH}Typ. = 3.4V$$

$$V_{out_LOW Typ.} = 0.2V$$

42. See Figure 22. Counter Circuit, If the clock frequency is 20KHz, find the counter's count sequence, modulus, and the MSB frequency.

Down Counter, sequence is 6 to 0

The Modulus is 7

The MSB Frequency is 2.857KHZ

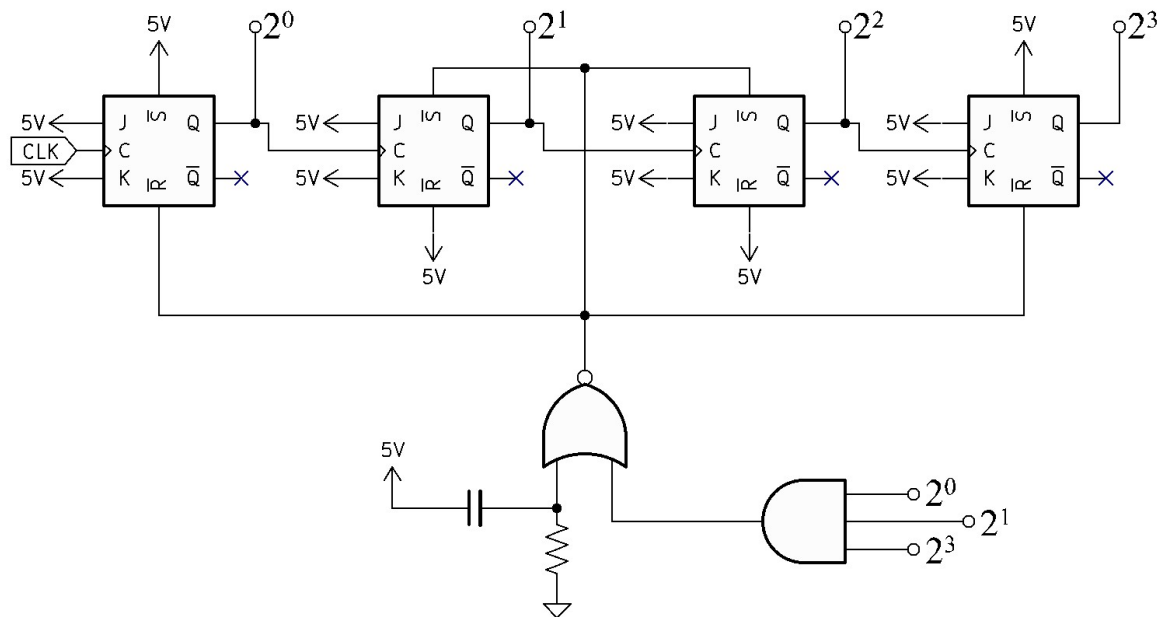


Figure 22: Counter Circuit

43. See Figure 23. Counter Circuit, If the clock frequency is 20KHz, find the counter's count sequence, modulus, and the MSB frequency.

Up Counter, sequence is 6 to 10

The Modulus is 5

The MSB Frequency is 4KHZ

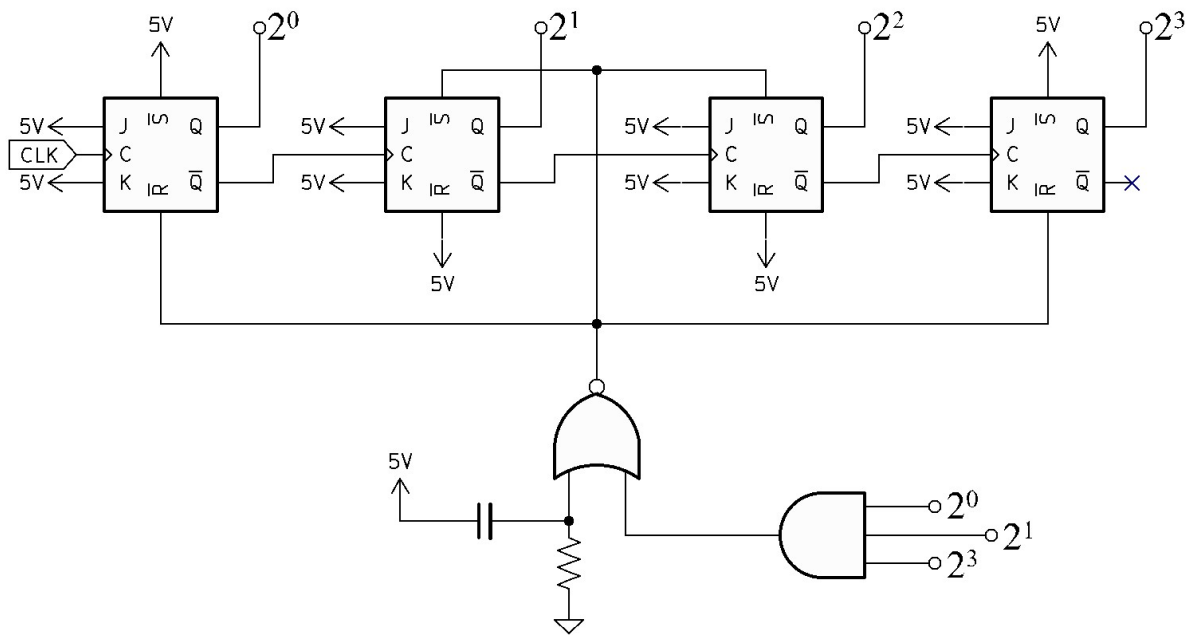


Figure 23: Counter Circuit